1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

Brand & Model: Intel(R) UHD Graphics 620
Type: Integrated

2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

DONE
https://youtu.be/JcQgZX-4W0o?si=w8_VbdLSBYe6QUtU

3. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

<u>Thermal Performance and Efficiency</u>
Air Cooling: Best in handling everyday task. Heat is directly removed from the system and thrown into the environment. Under heavy load, temperature increase may slow down your PC.
Water Cooling: It absorbs and transfers heat more efficiently from the system leading to consistently lower temperature. Hence it can be utilised for heavy gaming, overclocking or video rendering.

<u>Maintenance and Reliability</u>
Air Cooling: System is simple and highly reliable. If fan fails in working it is easily replaced as it is inexpensive.
Water Cooling: More system maintenance is needed, as water pump, hose connection and fluid should be checked over time and replaced.

<u>Cost</u>
Air Cooling: Affordable and cost effective compared to water cooling.
Water Cooling: It is significantly more expensive.

4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

When air coolers are used for streaming IPL matches or playing graphics-heavy games under intense load the fan will run at high speed producing considerable noise.
As the air coolers releases heat directly in the case it results in increasing graphic card baseline temperature.

Liquid coolers handles heavy game graphics and simultaneously dissipates heat resulting in lower noise levels, peak performance sustenance and less thermal throttling.

Hence, I choose liquid cooling system for the GPU.